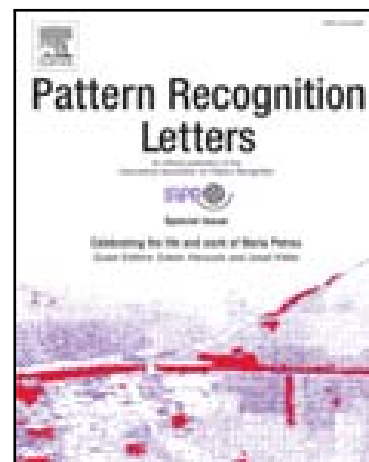


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A Novel Segmentation Technique for Online Handwritten Bangla Words

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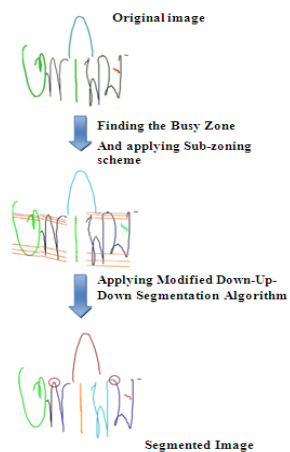
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Graphical Abstract

A Novel Segmentation Technique for On-line Handwritten Bangla Words

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A Novel Segmentation Technique for Online Handwritten Bangla Words

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ABSTRACT

In the present work, we have proposed a novel Bangla word segmentation technique that is based on stroke-level busy zone formation procedure. In an unconstrained domain, people often write text where strokes may be poorly aligned (due to multi-directional skewness) and varied combination of strokes with various types of joining between them are possible while forming the words. Hence, a segmentation approach for stroke extraction is pertinent for any stroke-based word recognition system. The presence of a large volume of symbols set (58 basic symbols with more than 280 compound characters) in Bangla script makes the task more challenging. In the current experiment, our stroke-level segmentation approach effectively handles such type of Bangla words. A sub-zoning scheme within busy zone followed by a modified Down- ζ Up- ζ Down (DUD) concept within these sub-zones has been used to find valid segmentation points. This scheme avoids over and under-segmentation issues caused by either inherent writing pattern or due to writing style variations up to certain extent. The proposed segmentation approach has been tested on 6,500 online handwritten Bangla word samples with 98.45% correct segmentation accuracy (compared with manually generated ground truth of the same database).

Keywords: Online Handwriting Recognition, Stroke segmentation, Stroke-based busy-zone, down- ζ up- ζ down strategy

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1. Introduction

Online Handwriting Recognition (OHR) is a challenging task due to the variability introduced by different individuals for their different writing styles. Things become more complicated in an unconstrained domain as people often write more than one stroke without lifting the pen. Along with that, there may be different types of joining patterns between the strokes. The segmentation technique plays an important role to break these joining and collect elementary strokes for further processing which involves recognition of those strokes and formation of words from the recognized strokes. A number of researchers such as Blumenstein and Verma (1999); Acharyya et al. (2013); Basu et al. (2002); Sarkar et al. (2008); Bhowmik

et al. (2012); Srihari (1993); Bozinovic and Srihari (1989); Yanikoglu and Sandon (1993); Chiang (1998); Gader et al. (1995); Han and Sethi (1995); Yanikoglu and Sandon (1998); Roy et al. (2005); Bhowmik et al. (2015); Singh et al. (2016); Karaa and Dey (2017); Karaa et al. (2016); Dey and Santhi (2017); Basu et al. (2009) have been investigated offline recognition of printed scripts like Bangla, English, etc. For online English word recognition, Shivram et al. (2013) have proposed a Conditional Random Field (CRF) driven beam search strategy that can be applied on a combined segmentation and recognition framework. To do so, firstly, authors have built a candidate segmentation lattice using over-segmented primitives of the word patterns and then recognition has been performed by synchronously matching lexicon words with nodes of the lattice. After integration of character recognition scores, geometric and spatial properties of the handwritten segments in a CRF model, probable search paths are measured. Authors have used

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beam search to minimize the set of likely search paths for efficient measurement. With the help of a special kind of directed labelled graphs, Vescovo and Rizzi (2007) have considered word sample as a sequence of strokes having knowledge of some inter-stroke information. Each node of the graph keeps feature information of a single stroke and edges stores the information about displacement between associated two strokes. Then employed graph classification system performs a special mining algorithm to perform different recognition tasks. Mori et al. (2014) have extracted global features for online character recognition whereas, Kumara et al. (2008) describes a large margin technique for writer independent online handwriting classification system.

While talking about online Bangla script, not only limited research documentations are available in the literature but also within those materials most of the works describe how to extract sub-stroke level features from word samples and recognition of the same. Fink et al. (2010) have segmented the strokes into sub-strokes and then eight scalar features representing shape, size and relative position have been computed. Then authors have proposed Hidden Markov Models (HMM) to recognize the target Bangla word from those sub-strokes. Bhattacharya et al. (2008) have proposed the analytic recognition approach that segments the input Bangla word into sub-strokes. They have recognized the segmented parts by designing a chain code histogram based feature vector. Finally, they have attempted to recognize the word sample by designing a module that uses a set of rules to construct the characters from strokes. Mo-hiuddin et al. (2011) have introduced a new hybrid classification strategy for the researchers of handwriting recognition domain. A Multi-Layer-Perceptron (MLP) architecture is used for feature extraction and SVM classifier is used for final classification purpose in this procedure. Sub-stroke level feature for the input word sample has been fed to the MLP trained by the backpropagation (BP) algorithm. The values estimated at the hidden nodes of MLP are treated as transformed feature vector and then fed to SVM as inputs. The technique of using MLP for feature estimation and SVM for classification, upgrades recognition accuracy from their individual level. Chowdhury et al. (2011) have described online Bangla word recognition by using a Weighted Finite-State Transducer (WFST) based language model. WFSTs may be treated as directed graphs whose edges are associated with input and output symbols with some weights. Unicode characters, graphemes, n-grams, or ligatures, etc. can be treated as symbols. In this paper, authors have investigated the role of WFSTs to model recognition lattice, lexicon as well as to edit the recognition hypotheses when none of the hypotheses corresponds to any word in the lexicon.

In the OHR domain, word recognition accuracy heavily depends on proper segmentation of word samples to obtain valid component strokes. A valid component stroke is one that belongs to any of the basic stroke classes. In the works described by Bhattacharya and Pal (2012); Bhattacharya et al. (2011); Ghosh (2009), authors have tried to collect stroke level information from target word by estimating busy zone over it and then applying the DUD approach for segmentation. To find busy zone of the word sample, an offline image is generated from

online information. A horizontal histogram is then computed in order to identify the busy zone boundary. Proper segmentation is heavily dependent on busy zone formation, which in turn depends on writing style and the nature of the skew of the word sample. If the skew is straightforward (either upward or downward) then skew detection and correction methodologies could be adopted easily, but when the skew is multi-directional (i.e. constituting strokes are at different levels of height) then busy zone formation really becomes complex and generates under-segmentation errors. Moreover due to inherent writing pat-

tern of characters like '৳', 'শ', DUD approach within busy zone causes certain over-segmentation issues. Ghosh (2009) has proposed a modified scheme for online Bangla word segmentation. Firstly, the word image is partitioned into two parts; upper zone and lower zone. Words are then segmented into a sequence of basic strokes by slightly modifying DUD concept. The author proposed to segment the word at the point where, there is a downward movement in upper zone and the slope of six consecutive pixels satisfies certain angular value.

Building of an online handwritten Bangla word recognition system using a stroke-based approach is not a much explored research area. Authors in Bhattacharya and Pal (2012); Bhattacharya et al. (2011); Ghosh (2009), have only tried to segment handwritten Bangla words to obtain constituent strokes from it. To the best of our knowledge no further work has been made in this area since 2012 either to overcome the problems (handling skew word samples and under/over segmentation issues) or to recognize the segmented strokes by adopting any suitable feature extraction strategy, that is the prior step to re-construct the words from the obtained segmented strokes.

In the present work, we have proposed a segmentation technique to obtain constituent basic strokes from online handwritten Bangla word samples and finally we have recognized those strokes. In this context, firstly, the strokes are categorized into three lists based on their positional information. We have proposed to form busy zone at stroke-level to handle multi-directional skewed words. Then a sub-zoning scheme within busy zone followed by a modified DUD concept within these sub zones has been used to find the segmentation points. The advantage and technical details of the stroke-level segmentation technique have been reported elaborately in the following subsections. We have also implemented a stroke recognition module to validate the proposed segmentation technique.

2. PROPOSED APPROACH

2.1. Segmentation of words

In this stroke based segmentation approach, strokes are firstly categorized into three groups based on their positional information. These groups are named as upper_zone, lower_zone and strokes_need_segmentation (SNS). The first two types are supposed to be free from segmentation for Bangla script but the SNS contains such type of strokes that need to be segmented. (The strokes in the SNS list contains multiple strokes with different types of joining between them due to writing without lifting up the pen). After categorization of the strokes, busy

zone is formed over the strokes belonging SNS stroke list. To do so, strokes are vertically divided into two equal halves and then for each half, Centre of Gravity (CG) has been computed. Angle made by line joining between two CGs and X axis describes the degree of skew for the stroke sample. This line is then shifted upward and downward up to certain height of the stroke to form the busy zone of the stroke sample. This process enables us to handle skewed strokes. A modified DUD approach is then adopted for segmentation purpose by dividing the stroke-level busy zone into different sub-zones. Up to a certain limit, the sub zoning scheme helps to handle different joining patterns between strokes that are not aligned horizontally and finds the valid segmentation points. This scheme is

also useful for the characters like 'ল', 'শ', etc. whose inherent writing pattern may mislead DUD approach causing over-segmentation. Algorithm 1 describes the overall procedure of the proposed segmentation algorithm.

We have categorized the strokes into three lists as described in step 1 of Algorithm 1. It is clearly seen from Fig.1 (a), that strokes #2, #3 and #5 should be included into lower_zone list that need no more segmentation. Whereas, strokes #1 and #4 should be included into SNS because they contain multiple connected strokes written in a single stretch (written without lifting up the pen) and thereby those need to go through some efficient segmentation process. Algorithm 2 describes the process of strokes categorization and Fig. 1(b) pictorially describes this algorithm for first three strokes of Fig.1(a).

Algorithm 1: Overall steps required for stroke based segmentation

1. Divide all the strokes into three stroke lists; upper_zone, lower_zone and SNS using Algorithm 2.
 2. For strokes $i=1$ to N from SNS do,
 3. Calculate busy zones for strokes using Algorithm 3.
 4. Apply modified DUD approach within stroke-level busy zone for segmenting them into basic strokes.
-

During the execution of Algorithm 2, step 3.2 finds the set of strokes s_k within the width of s_i . For example, as shown in Fig. 1, when stroke #1 is considered then stroke #2 and #3 are also examined because these two strokes fall within the width of stroke #1. Due to nature of Bangla script it can be said that within $\{s_k + s_i\}$ one stroke should be major that needs to go through segmentation process and others may belong to upper_zone or lower_zone list or both that need not be segmented. Step 3.3 finds the major stroke s_t on the basis of pixels count from the set of overlapping strokes $\{s_k + s_i\}$. The major stroke is the one which contains maximum number of pixels. We have set this criteria because ascendant (upper_zones) and descendant (lower_zones) strokes generally contain less number of pixels than strokes in the middle zones (position between upper_zone and lower_zone) of a word.

Algorithm 2: Steps for stroke categorization

1. For all strokes $i=1$ to N
 2. $flag_i=0$
 3. For all strokes s_i where $i=1$ to N
 - 3.1 if $flag_i=0$ then do
 - 3.2 Find the set of strokes $\{s_k\}$ that contributes some pixels within the width of s_i
 - 3.3 Find stroke s_t from set of strokes $\{s_k\}$ and s_i that contains maximum number of pixels
 - 3.4 Place s_t in SNS and set $flag_t=1$
 - 3.5 For each stroke s_p from set of strokes $\{s_k + s_i - s_t\}$ do,
 - 3.5.1 Check if CG of s_p lies top of minY and maxY of s_t then
 - 3.5.1.1 Place stroke s_p in upper_zone list and set $flag_p=1$
 - 3.5.2 Check if CG of s_p lies below minY and maxY of s_t then
 - 3.5.2.1 Place stroke s_p in lower_zone list and set $flag_p=1$
-

In Fig. 1, stroke #1 qualifies as SNS because of having maximum number of pixel points. Then, CGs of all overlapping strokes #1, #2 and #3 have been computed that are termed as G1, G2 and G3 respectively. In Fig. 1, d1, d2, d3, d4 denote the distance measures from G2 and G3 to maxY and minY respectively of stroke #1. As d1 and d3 both have greater values than minY, as well as d2 and d4 have greater values than maxY of stroke #1, hence both the strokes #2 and #3 lie below the stroke #1 and thereby can be placed in the lower_zone. Similarly between strokes #4 and #5, first one goes into SNS whereas, later one falls into lower_zone.

After getting SNS, the busy zone has been formed by adopting Algorithm 3 into enter to the segmentation process. This helps to get rid of estimating busy zones for SNS which are poorly aligned due to writing variations of individuals.

Fig. 2(a-c) describes the pictorial view of busy zone formation for strokes #1 and #4 from Fig. 1(b). Fig. 2(a) reflects the vertical division of strokes #1 and #4 based on calculated CGs. Fig. 2(b) depicts the CGs for two sub-parts that are connected by drawing a line between them and the angle theta (θ) made by this line with x axis denotes the skew of the stroke. In this current experiment, detected skew is not corrected but this line is shifted in upward and downward directions (up to Top_Line and Bottom_line as shown in Fig. 2(c) by taking the help of computed vertical histogram of the sample stroke) to form the busy zone for the stroke sample. This strategy minimizes the time complexity by reducing the extra calculation for skew correction. Again, as this is a stroke based word segmentation approach, hence poor alignment of strokes has minimal impact on the final segmentation of the words even if the words are multi-directionally skewed.

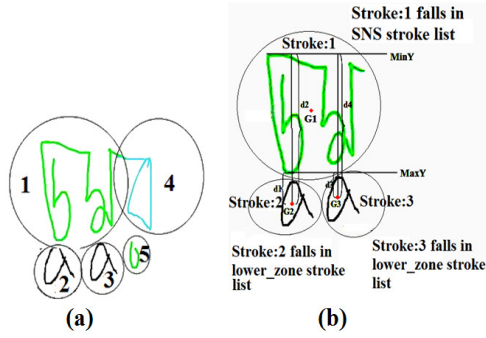


Fig. 1. (a) Strokes of a sample word according to writing order, (b) stroke categorization using Algorithm 2

Algorithm 3: Steps to form stroke-level busy zone

1. Vertically divide a stroke into two parts.
2. Calculate CG for both the left and right parts.
3. Move the line joining two CGs up to Top_Line and Bottom_Line to form busy zone of the stroke

After forming stroke-level busy zone, the modified DUD concept has been adopted for stroke segmentation purpose to yield the required component strokes. Before adopting this concept, a minor change has been made in the area within Top_Line and Bottom_Line (as shown in Fig. 3(a)). This zone is divided into five sub-zones to cope with the variability due to different structural patterns of strokes and also the writing variations of individuals. These sub-zones has been named as Zone1 (Top_Line to Top_Line+t1), Zone2 (Top_Line+t1 to Top_Line+t2), Zone3 (Top_Line+t2 to Top_Line+t3), Zone4 (Top_Line+t3 to Top_Line+t4) and Zone5 (Top_Line+t4 to Bottom_Line) as shown in Fig. 3(a). Here, t1=height of busy zone/6, t2= height of busy zone/3, t3=height of busy zone/2 and t4= height of busy zone*3/4 where height of busy zone= Bottom_Line - Top_Line.

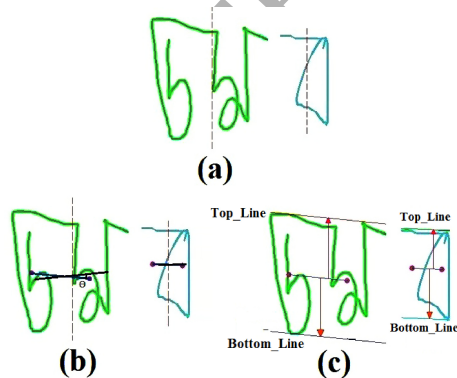


Fig. 2. (a) Vertical division of SNS strokes into two parts (b) measurement of skew angle (c) formation of busy zone

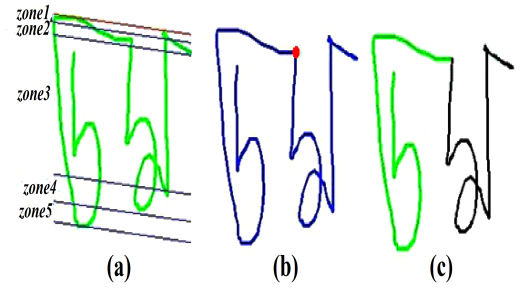


Fig. 3. (a) Division of busy zone into sub-zones, (b) identification of valid segmentation point, and (c) component strokes after segmentation

As the joining between characters in Bangla words are found in the matra region, hence we assume that segmentation takes place in the Zone2 only. After forming sub-zones, the path/positions of all the pixel points (according to writing sequence) are named by either "up" or "down" on the basis of relative positional order of different sub-zones. The convention for the direction of "up" to "down" movement used here is shown in the Fig. 9(b) and reverse direction is true for "down" to "up" movement. For example, if a pixel goes from Zone2 to Zone3 then it is named as "down", whereas, when pixel goes from Zone3 to Zone2 then it is named as "up" and pixels within the same region get the same naming convention. Initially it starts by "donot know" status. Then depending on writing style of the stroke, pixels are named as "up" or "down" by looking at the movement of the pattern through different sub-zones. Now we segment the strokes (having multiple connected components) in Zone2 region when it tries to go downward, i.e. towards Zone3, after checking the movement over different sub-zones from start or previous segmentation point of the stroke to the point of consideration. If this movement satisfies any of the following five conditions (described from case 1 to case 5) then only segmentation takes place. Fig. (4-8) depicts correct segmentation outcomes for these five cases respectively. Red marked circles highlights the joining between strokes.

Case 1: Movement starts from Zone1 or Zone2, and then follows the sequence

Zone3-;Zone4-;Zone3-;Zone2/(Zone2-;Zone1-;Zone2)-;Zone3

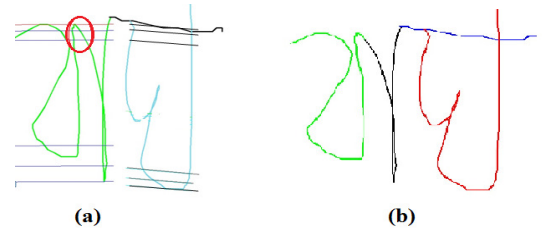


Fig. 4. (a) Stroke sample follows pattern movement (movement of pixels) as mentioned in case 1 (b) Segmented elementary strokes (elementary strokes are represented by different colours)

Case 2: Movement starts from Zone1 or Zone2, and then follows the sequence Zone3-;Zone4-;Zone5-;Zone4-;Zone3-;Zone2/(Zone2-;Zone1-;Zone2)-;Zone3

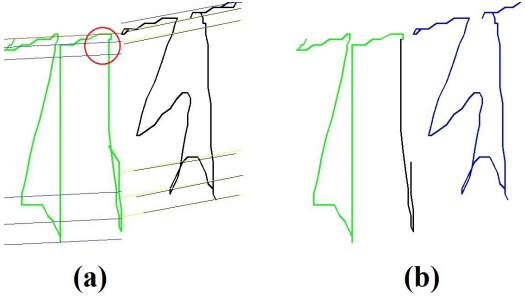


Fig. 5. (a) Stroke sample follows pattern movement (movement of pixels) as mentioned in case 2 (b) Segmented elementary strokes (elementary strokes are represented by different colours)

Case 3: Movement starts from *Zone3* and follows the sequence *Zone4-;Zone5-;Zone4-;Zone3-;Zone2-;Zone3*

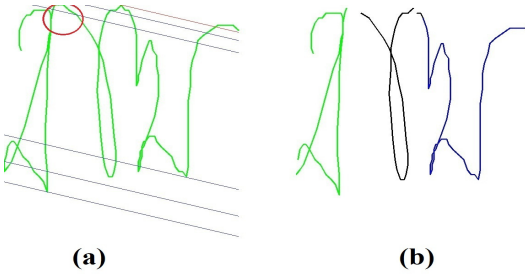


Fig. 6. (a) Stroke sample follows pattern movement (movement of pixels) as mentioned in case 3 (b) Segmented elementary strokes (elementary strokes are represented by different colours)

Case 4: Movement starts from *Zone3* and follows the sequence *Zone4-;Zone5-;Zone4-;Zone3-;Zone2-;Zone1-;Zone2-;Zone3*

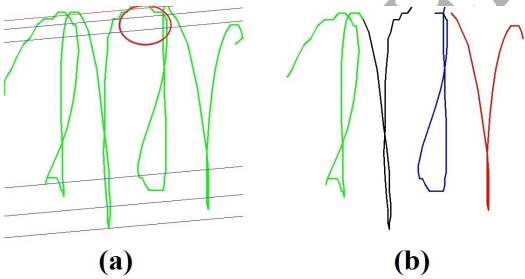


Fig. 7. (a) Stroke sample follows pattern movement (movement of pixels) as mentioned in case 4 (b) Segmented elementary strokes (elementary strokes are represented by different colours)

Case 5: Movement starts from *Zone3* and follows the sequence *Zone4-; Zone3-;Zone2-;Zone1-;Zone2-;Zone3*

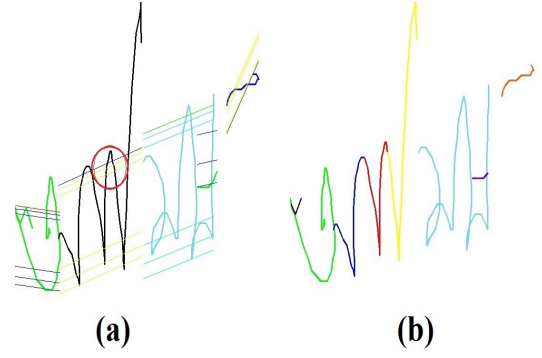


Fig. 8. (a) Stroke sample follows pattern movement (movement of pixels) as mentioned in case 5 (b) Segmented elementary strokes (elementary strokes are represented by different colours)

Fig. 3(b) reflects the positions of segmentation points after executing this modified segmentation scheme for stroke #1 shown in Fig. 2(c). Clearly stroke #1 joins two different symbols from basic alphabet set. The pattern movements of pixels for this stroke satisfy case 4 and hence it gets segmented at position shown in Fig. 3(b). Fig. 3(c) reflects the obtained component strokes after segmentation.

Fig. 9(a-b) highlights the advantage of using sub-zoning scheme with modified DUD approach which is applied on two different word samples belong to SNS. Fig. 9(a) displays the pixel contributions of a stroke (shown in black colour) within different sub-zones that contains two different isolated stroke symbols and depicts the faced complexity as well as the advantage of using this scheme for proper segmentation. The joining between the symbols is happen in such a way that they lie at different heights, thus making some skew. The proposed busy zone formation technique works well in this case too. From the figure it can be observed that second part of the stroke resides quite below of the TOP_LINE which could have been misled the segmentation technique causing under-segmentation error. But the sub-zoning scheme along with modified DUD concept prevents such issues and helps to locate the proper segmentation point. From Fig. 9(a), it is observed that the stroke satisfies Case 3 and thus gets segmented properly.

In Fig. 9(b), the writing pattern of stroke 'ल' starts from *Zone3* and after crossing *Zone4* it enters into *Zone1* and tends to move towards *Zone3*. Clearly, writing pattern follows a down-up-down movement inside *Zone2* (see Fig. 9(b)). But the stroke is not eligible for segmentation, because the previous stroke has not been traversed through *Zone5* and thereby it does not satisfy any of the conditions mentioned before.

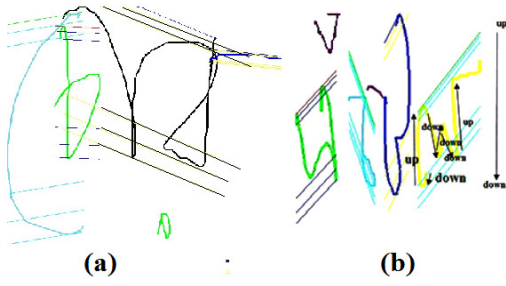


Fig. 9. (a-b) Sub-zoning scheme together with modified DUD approach to handle tricky stroke patterns for two different types skewed words

The examples of the steps involved in the segmentation module for a skewed word sample are shown in Fig. 10(a-e).

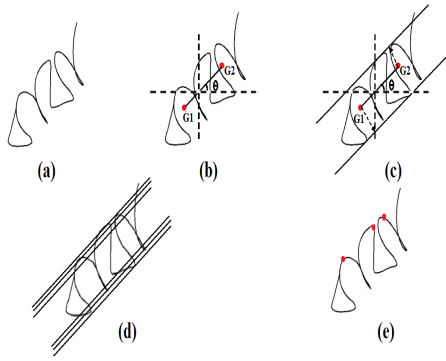


Fig. 10. (a) Skewed handwritten Bangla word sample (b-c) skew detection and busy zone formation using Algorithm 3 (d) formation of sub-zones within busy zone (e) achieved segmentation points after applying modified DUD approach within sub-zones.

2.2. Pre-processing and Feature extraction

After passing through the segmentation process, we have manually analyzed the obtained strokes and grouped them into 59 classes [Roy (2012)]. Existence of some nearly similar shape structures has been observed during statistical analysis of the stroke database considered here. In order to differentiate such similar shaped patterns, some local information are estimated to produce discriminating features for the recognition purpose. Stroke pre-processing and modified distance based feature extraction technique for Bangla stroke and character recognition were presented in Sen et al. (2015, 2016). According to this approach, strokes are firstly divided into N hypothetical segments. Feature values are estimated by the distances measured from each point to every other points of the stroke. Following the works Sen et al. (2015, 2016), we set the value of N to 16. Fig. 11(a-c) reflects feature calculations for three points. For a N -segmented stroke, this approach produces $N*(N+1)/2$ element feature vector. Thereby, 136-element feature vectors have been produced as an outcome for 16-segmented stroke and same are fed to several classifiers. In this current experiment, in total 13,630 stroke samples have been considered for the recognition purpose.

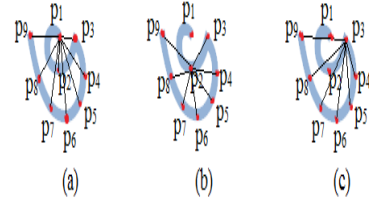


Fig. 11. Distance based feature calculation while considering (a) first point (b) second point and (c) third point for a Bangla character constituted with 9 points

3. Result and Discussion

In this current experiment, 6,500 online handwritten Bangla word samples have been used to verify the efficiency of the proposed segmentation algorithm. The benefits of this approach are 3-fold:

- 1) Usage of available online stroke information to make the segmentation process effective for multi-directional skewed word sample (as shown in Fig. 12(a)), where constituent strokes are poorly aligned, and for samples like in Fig. 12(b) where almost every character contains modifier in the lower part. Looking at the multi-directional skew in Fig. 12(a) and structural pattern of Fig. 12(b), this can be understood that busy zone formation for the whole word by following Bhattacharya and Pal (2012); Bhattacharya et al. (2011) may be erroneous, and which in turn may result either over or under-segmentations. Whereas, in the proposed approach, main focus is on stroke, rather than the entire word. Hence, the issue of skewness become easier to deal with as stroke-level local skew is much lesser than the word-level global skew. Along with that, stroke categorization technique helps to identify only those strokes, which need to be segmented, having multiple connected components.
- 2) Our proposed busy zone formation over target strokes avoids



Fig. 12. Examples of words for which busy zone formation by computing horizontal histogram may mislead segmentation process. The reason for (a) is skew and for (b) every character contains a symbol in its lower part

skew correction process as discussed earlier. Hence, this approach reduces execution time needed for the skew correction before entering segmentation process.

- 3) Division of busy zone into sub-zones followed by modified DUD technique, not only reduces the time complexity/difficulty of verifying different constraints for candidate segmentation point selection introduced in Bhattacharya and Pal (2012); Bhattacharya et al. (2011), but also helps to handle different types of handwriting styles in an unconstrained domain.

The effectiveness of proposed segmentation technique is evaluated by its prediction ability to give a correct classifica-

tion of events as an attack or a normal behavior. According to the real nature of a given event and the prediction, four possible outcomes may be there. **True negatives and true positives correspond to correct operations observed. True negatives (TN) are events that are actually normal and also successfully labeled as normal, events those are actually attacks and are successfully labeled as attacks are termed as true positives (TP). Similarly, false positives (FP) corresponds to normal events but classified as attacks and false negatives (FN) are actually attack events but incorrectly classified as normal events.** Hence, in our case, TP, FP and FN reflect correctly segmented, over-segmented and under-segmented points respectively. In the proposed approach the value of TN is absent.

From the generated values of TP, FN and FP, precision ($P = TP/(TP + FP)$), recall ($R = TP/(TP + FN)$) and F-measure ($F = 2 * ((P * R)/(P + R))$) of the proposed system are computed. Table 1 describes the detail segmentation results of the proposed scheme.

Table 1. Detail segmentation results of online handwritten Bangla words

Observation Parameters	Observed Outcome
Ideal Segmentation points	3930
Correctly Segmented Points (TP)	3869 (98.45%)
Under-Segmented Points (FN)	61 (1.55%)
Over-Segmented Points (FP)	72 (1.83%)
P	0.98
R	0.98
F	0.98

Some correctly and incorrectly segmented word samples are shown in Fig. 13(a-f) and Fig. 14(a-d) respectively. Looking at all the incorrectly segmented words samples, it is found that system produces few over-segmentation and under-segmentation cases. After a close inspection, it has been observed that the word samples of Fig. 14(a-b) are the examples where not only constituent strokes are poorly aligned (horizontally) due to inter-stroke height difference but also some strokes have intra-stroke differences (i.e. within that stroke multiple connected strokes are there at different level of height). Hence, busy zone formation over such type of strokes misleads the segmentation process. Whereas, the images of Fig. 14(c-d) are very rare examples, found in the database, which are over-segmented by the proposed technique. Fig. 15(a-c) illustrates the formation of busy zones for the strokes produce either over or under-segmentation errors. Analysis of improper segmentation reveals that, few over-segmentation issues arise in case

of characters ল and দ . From the shapes of the characters and their common variants while writing, it can be perceived that both have a natural shape which contains a curve in the upper zone which comes from Zone4 or Zone5 and reaches closer to the matra region which is the possible segmentation zone. Along with the inherent nature of the stroke, writing variations of individuals may also increase the chances of over-segmentation. If those characters have a substantial part after this possible segmentation zone, then this technique generates some extra segmentation points. This over-segmentation issues

can be taken care of at the time of recognition/post processing phase. In the current work, we have minimized the under-segmentation issues in most of the cases except the cases where extreme writing variations are found.

We have also investigated the segmentation outcome by adopting the procedures described in Bhattacharya and Pal (2012) and Bhattacharya et al. (2011). In this case, we have achieved 97.65% correct segmentation accuracy with 17% over-segmentation and 2.35% under-segmentation rate, which is quite lower than the proposed scheme. We have measured the computation time of the proposed segmentation technique and it is approximately 0.6 second per word sample. We have also investigated the computational time for the other two works by Bhattacharya and Pal (2012), Bhattacharya et al. (2011). The computation time for these work are 0.85 second and 0.94 second respectively. Please note that the experiment is done on a machine whose specification is as follows: Intel processor with 2.16 GHz processing speed and 4 GB RAM.

Table 2 describes detail recognition results of a total of 13,630 stroke samples when distance based feature has been applied. Some well known classifiers such as MLP, C-SVC type SVM with default kernel (Radial basis function) and BayesNet have been employed for stroke recognition purpose. From bold-faced entries in Table 2, it can be observed that SVM outperforms other classifiers by showing highest stroke recognition accuracy of 92.95%.

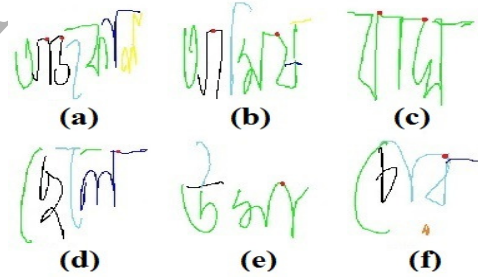


Fig. 13. (a-f) Examples of sample words which are correctly segmented

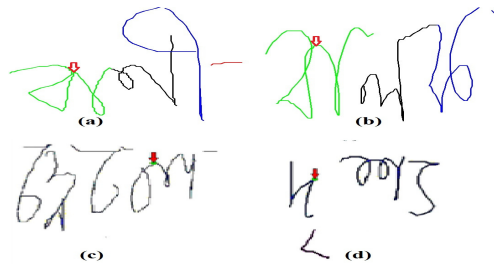


Fig. 14. Sample words which are not segmented correctly: (a-b) cases of under-segmentation and (c-d) cases of over-segmentation. Arrows indicate the positions where under-segmentation or over-segmentation has occurred

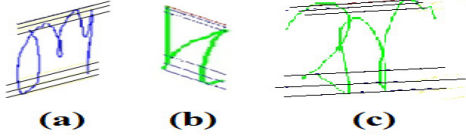


Fig. 15. Busy zones for the stroke samples produce either over (a-b) or under-segmentation (c)

Table 2. Success rates of different classifiers for the recognition of strokes of the online Bangla words with distance based feature

Stroke database size (#)	Classifier used	Recognition accuracy (in %)
13630 (59 Class)	MLP	92.65
	SVM	92.95
	BayesNet	86.63

As in the preliminary experimentation of stroke recognition, C-SVC type SVM shows the best result among other classifiers, so we have decided to test more with varying the parameters of SVM in order to enhance the performance of the same. Table 3 reports result observed here. **Three kernel variants of SVM are applied in the experiment. Given vectors u, v the polynomial, Gaussian and linear kernel kernels are defined through**

$$k(u, v) = (\gamma * u^t * v + \alpha)^d$$

$$k(u, v) = \exp(-\beta * \|u - v\|^2)$$

$$k(u, v) = u^t * v$$

where α, β, γ, d are real numbers and $d, \gamma > 0$. d is the degree of polynomial kernel. From Table 3 it can be seen that polynomial kernel of SVM gives best classification result when all the tuning parameters are set with default value (e.g. the default value of d is 3).

Table 3. Stroke recognition results with C-SVC type SVM for three different kernels

Kernel variants of SVM	Recognition accuracy
Gaussian	92.95%
Polynomial	95.33%
Linear	94.95%

Table 4. Stroke recognition results produced by SVM (with polynomial kernel) varying the *epsilon* and *gamma* values

SVM Type	Kernel type	epsilon	gamma	Recognition accuracy (in %)	Model building time (in sec)
C-SVC	Poly-nomial	0.001	0.0	95.33	23.56
		0.002	0.0	95.32	48.81
		0.003	0.0	95.32	50.45
		0.005	0.0	95.32	50.47
		0.001	0.1	95.01	65.75
		0.001	0.2	94.99	67.62
		0.001	0.3	94.96	69.14

We have also investigated the accuracy of the SVM by tuning the parameters *epsilon* and *gamma* for polynomial kernel. *epsilon* controls tolerance for termination criteria and *gamma* tunes the applied kernel function. The detailed information regarding recognition accuracy and time taken to build the model are mentioned in Table 4. Here, we have considered two situations for tuning the parameters. In the first case, we have fixed the value of *gamma* and increased the *epsilon* values gradually. Consequently, we have observed that either stroke recognition accuracy decreases or model building time increases with increasing *epsilon* value. In the second case, the value of *epsilon* is kept fixed and gradually we have increased the *gamma* values. In this situation, model building time increases and recognition accuracy also gets decreased with the increase of *gamma* value. Gray cell of Table 4 highlights the situation where best recognition rate along with minimum model building time is observed (when we have set the values of *epsilon* and *gamma* to 0.001 and 0 respectively).

Some misclassified stroke pairs along with their misclassification rates are shown in Table 5. Deeply analyzing the misclassified stroke pairs, it can be said that strong structural similarity is the sole reason for the misclassification.

We have also implemented 64-dimensional direction feature vector used by the authors in Bhattacharya and Pal (2012); Bhattacharya et al. (2011) for stroke recognition purpose and obtained classification accuracy as 94.2%.

Table 5. Some confusing stroke pairs

Misclassified stroke pairs	# of Misclassification (# of total samples)
	24 (490)
	72 (484)
	65 (410)
	30 (470)

We have also compared the efficiency of the proposed segmentation approach with two previous techniques viz.

Bhattacharya and Pal (2012); Bhattacharya et al. (2011) related to online Bangla word segmentation. Table 6 highlights the comparison details. From the records of this table it can be stated that proposed system outperforms the others. From Table 6 it can be observed that our technique minimizes the chances of both under and over-segmentation issues while experimenting on a larger volume of database. In an unconstrained domain, the word samples may be poorly aligned with multi-directional skew. In such cases busy zone formation may be erroneous and leads to improper segmentation of the words, because DUD approach works only within busy zone to identify segmentation points. To overcome this situation, we have introduced stroke based busy zone formation strategy which manages inter-strokes height differences effectively. According to segmentation techniques described in Bhattacharya and Pal (2012); Bhattacharya et al. (2011) maximum

samples of 'ন', 'ল', 'দ' and sometimes 'ছ' may get over-segmented due to their inherent shape structure. Because the said patterns follow a DUD movement within the busy zone and thus get segmented. To prevent over-segmentation, though authors in Bhattacharya and Pal (2012); Bhattacharya et al. (2011) have introduced some validation checking but in certain cases it fails. The proposed sub-zoning scheme within busy zone followed by modified DUD concept lessens over-segmentation issues by introducing above mentioned five cases. Again, if we try to analyze the cause of under-segmentation as in Bhattacharya and Pal (2012); Bhattacharya et al. (2011), then it comes out that the nature of joining between multiple valid stroke symbols written within a single stroke makes the errors happen. This scheme is also useful to handle under-segmentation issues up to certain extent as shown in Fig. 9(a). The detail comparison of the segmentation results from Table 6 reflects the truthfulness of the above facts.

Table 6. Comparison of proposed segmentation approach with some past works

Comparable Attributes	Bhattacharya and Pal (2012)	Bhattacharya et al. (2011)	Proposed Work
Databasesize (#words)	2000	2000	6500
Ideal segmentation points	2698	2698	3930
Correctly segmented points(TP)	2641 (97.89%)	2635 (97.67%)	3869 (98.45%)
Under Segmented points (FN)	57 (2.11%)	63 (2.33%)	61 (1.55%)
Over Segmented points (FP)	532 (16.47%)	532 (16.47%)	72 (1.83%)
Precision	83.23%	83.20%	98.17%
Recall	97.89%	97.66%	98.45%
F-measure	89.91%	89.87%	98.29%

Table 7. Comparison of proposed feature extraction approaches with some past works

Comparable Attributes	Bhattacharya and Pal (2012)	Bhattacharya et al. (2011)	Proposed Work	
Stroke Database	2000	2000	6500	
Used Feature	Direction Based	Direction Based	Direction Based	Distance Based
Stroke Recognition Accuracy	97.68%	97.67%	94.2%	95.33%

Table 7 highlights the detailed comparison of the proposed system with works described by Bhattacharya and Pal (2012); Bhattacharya et al. (2011). From this table, it can be said that stroke recognition accuracy of the present technique is comparable with state-of-the-art. In this present experiment 13,630 stroke samples are considered for the recognition using distance based feature and we have obtained 95.33% accuracy, which is quite good in comparison with produced success rate (94.2%) for directional feature. Though the result is lesser than the experiments done by authors in Bhattacharya and Pal (2012); Bhattacharya et al. (2011), but for the current experiment we have considered larger database than the said techniques. Hence, more error is inevitable as database includes larger stroke variations.

4. Conclusion

To design a successful OHR engine, segmentation of words into strokes is a pre-requisite. The more accurate is the segmentation module, the less error-prone is the OHR engine. In this current work, we have developed a segmentation approach which can handle almost all type of Bangla words from a simple horizontally aligned to multi-directional skewed ones. Though some of the skewed words where component strokes contain multiple connected parts are not segmented properly but considering the complexities of large variations in the handwriting, it can be said that this approach performs reasonably well. Handling extreme variations in writing styles of online Bangla handwritten words and building effective feature extraction technique for the stroke recognition module can be thought of as future scopes of the current work.

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